Executive Healthcare Analytics

DATA QUALITY ASSESSMENT (DQA) REPORT

Target System: Executive Operations & Financial Dashboard

Primary Metric Suites: Encounter Volumes & Claim Expenses

Operational Color Palette: Deep Teal (#005F73)

Financial Color Palette: Maroon (#800020)

1. Executive Summary & Assessment Scope

This Data Quality Assessment (DQA) establishes the technical audit framework for the Executive Healthcare Operations & Financial Dashboard. The objective of this evaluation is to verify data integrity across operational datasets (patient encounters, clinical visits) and financial datasets (claim costs, payer coverage payouts) before executive deployment. Ensuring strict data hygiene directly protects executive decision-making across volume dynamics and revenue allocation.

2. Data Quality Dimensions Audit

A. Completeness Audit

Evaluates whether all mandatory primary keys, temporal identifiers, and financial amounts exist without structural omission.

- **Encounter Keys:** Verified zero null records in `ENCOUNTER_ID`, `PATIENT_ID`, and `START` timestamp fields.
- **Financial Attributes:** Confirmed 100% population for `TOTAL_CLAIM_COST` and `PAYER_ID` across all claim rows to ensure non-distorted summary totals.

B. Accuracy & Validity

Guarantees that operational workflows adhere to clinical and financial business rules.

- **Temporal Logic:** Audited visit timelines to ensure discharge/stop timestamps strictly follow admission/start timestamps (`STOP ≥ START`).
- **Financial Bounds:** Validated that `TOTAL_CLAIM_COST` and `PAYER_COVERAGE` contain non-negative numeric values (≥ $0.00).

C. Referential Consistency

Confirms valid relational linkages between fact tables and dimension tables.

- **Patient Integrity:** Every encounter record maps directly to an existing entry in the Patient Master dimension.
- **Payer Integrity:** Confirmed zero orphan records between historical claims and the Payer dimension table, preventing unclassified categories in the Payer Cost Distribution donut chart.

3. Quality Assessment Matrix

Quality Dimension	Validation Rule & Target	Affected Visual / KPI Card	Status
Completeness	Null check on ENCOUNTER_ID & START (100% Populated)	Total Encounters KPI Card	Passed
Uniqueness	Primary key duplicate check on ENCOUNTER_ID (0 Duplicates)	Encounters Trend Over Time	Passed
Accuracy	Duration non-negative constraint (STOP ≥ START)	Average Encounter Duration KPI	Passed
Consistency	Foreign Key link check to Payer Master Table (0 Orphans)	Payer Cost Distribution (Donut)	Passed
Financial Validity	Non-negative claim values (TOTAL_CLAIM_COST ≥ 0)	Average Total Claim Cost KPI	Passed

Governance Note: All verified data points feed directly into the top horizontal slicers (Date Range, Care Setting / Class, and Payer), ensuring real-time cross-filtering across both operational (Teal) and financial (Maroon) analytics components.